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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Seventh Semester B.E. (CCE) (Credit System) Degree Examinations

Supplementary Examinations - May 2025

21CC504 / 20CC403 - PRINCIPLES AND PRACTICES OF SOFTWARE ENGINEERING

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

		Unit – I			
		Marks	BT*	CO*	PO*
1.	a) Explain the four important attributes which all software products should have.	5	L*2	1	2
	b) With the help of block diagram and activity model explain an insulin pump control system. Also comment on the essential high level requirements that this system must meet.	10	L2	1	2
	c) Bring out the difference between generic and bespoke software.	5	L2	1	2
2.	a) Explain Incremental development process model with a neat block diagram. List its benefits and problems.	10	L2	2	2
	b) With the suitable block diagram explain waterfall model.	6	L2	2	2
	c) Explain the types of software process models.	4	L2	2	2
3.	a) Illustrate requirement engineering process with a neat block diagram.	10	L2	2	2
	b) Explain requirement elicitation and analysis process and briefly explain ethnography technique.	10	L2	2	2
		Unit – II			
4.	a) Explain event-driven model with a state diagram of microwave oven application.	10	L2	3	2
	b) Draw and explain a context model for patient information system.	6	L3	3	2
	c) Explain use case models with example.	4	L2	3	2
5.	a) Define sequence models. Draw and explain the sequence diagram for describing weather data collection.	8	L3	3	2
	b) Define state diagram. Draw and explain weather station state diagram.	8	L3	3	2
	c) Mention and explain the weather station object classes.	4	L3	3	2
6.	a) Describe how SCRUM process helps in agile project management.	10	L2	4	2
	b) Write a note on pair programming.	5	L2	4	2
	c) Write a short note on refactoring.	5	L2	4	2
		Unit – III			
7.	a) Define Risk management. Explain in detail risk management process with block diagram.	10	L2	5	2
	b) Define project scheduling. Explain the project scheduling process and its activities with the block diagram and example.	10	L2	5	2
8.	a) Explain the software review process.	10	L2	5	2
	b) Explain with block diagram, key stages in product measurement process.	10	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
